

Probability Theory Exam Cheat Sheet

Your Ultimate Study Resource

September 13, 2026

1 Sample Spaces & Language Traps

The sample space (S) is the set of all possible outcomes. Always use these three filters to interpret word problems correctly:

- **Order Matters:** Look for phrases like "*with respect to birth order*" or "*in order*". This means $(B, G) \neq (G, B)$.
- **Can Things Repeat?** Look for "*without replacement*" (items countdown, cannot repeat) vs. "*with replacement*" (items reset, can repeat).
- **Counted vs. Measured:** Counted scenarios are discrete sets (e.g., $S = \{2, 3, 4, \dots\}$). Measured scenarios are continuous intervals (e.g., $S = [0, \frac{1}{3}]$).

1.1 Key Definitions

- **Exhaustive Events:** A set of events that completely covers the entire sample space. At least one of them **must** happen.

$$A_1 \cup A_2 \cup \dots \cup A_n = S \implies P(A_1 \cup A_2 \cup \dots \cup A_n) = 1$$

- **Mutually Exclusive Events:** Events that cannot physically happen at the same time ($A \cap B = \emptyset$). They have zero overlap: $P(A \cap B) = 0$.
- **Independent Events:** The occurrence of one event has absolutely no effect on the probability of the other.

2 Core Probability Theorems

2.1 1. The Addition Theorem (The "OR" Rule)

Used to find the probability that **at least one** of two events happens.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Edge Case (Mutually Exclusive): If they cannot overlap, drop the subtraction part:

$$P(A \cup B) = P(A) + P(B)$$

2.2 2. Conditional Probability

Calculates the likelihood of an event happening **given that** another event has already occurred. The term to the right of the vertical bar ($|$) is a guaranteed past fact.

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

2.3 3. The Multiplication Theorem (The “AND” Rule)

Used to find the probability of a sequence of events happening together over a timeline.

$$P(A \cap B) = P(A) \times P(B | A)$$

Edge Case (Independent Events): If the events do not affect each other, the formula simplifies to:

$$P(A \cap B) = P(A) \times P(B)$$

3 Bayes’ Theorem & The Tabular Method

Bayes’ Theorem updates the probability of a cause (Hypothesis H_i) after an effect (Evidence E) has occurred.

$$P(H_i | E) = \frac{P(H_i) \times P(E | H_i)}{\sum P(H_i) \times P(E | H_i)}$$

3.1 The Bayes’ Table Shortcut

Instead of dealing with the large fraction, lay your data out in a systematic table. Work from left to right, column by column:

Hypothesis (H_i)	Prior Prob. $P(H_i)$	Conditional Prob. $P(E H_i)$	Joint Prob. (Col 2 $\times$ Col 3)
H_1	Given value	Given value	$P(H_1) \times P(E H_1)$
H_2	Given value	Given value	$P(H_2) \times P(E H_2)$
Total	Must equal 1	<i>Do not add</i>	Total Joint = $P(E)$

4 The Ultimate Exam Shortcuts

4.1 1. The Complement Trick (Focus on Failure)

Whenever a question asks for the probability that a problem “*gets solved*” or happens “*at least once*” across independent paths, calculate total failure first, then subtract it from 1.

$$P(\text{Success}) = 1 - P(\text{Total Failure})$$

$$P(\text{Success}) = 1 - [P(A \text{ Fails}) \times P(B \text{ Fails}) \times P(C \text{ Fails})]$$

4.2 2. Parallel Path Modeling (Contradiction Puzzles)

When tracking scenarios where individuals disagree or cross paths:

1. Map out all winning individual sequences using the **Multiplication Theorem**.
2. Add those distinct pathways together using the **Addition Theorem**.

$$P(\text{Contradiction}) = P(A \text{ Truth} \cap B \text{ Lie}) + P(A \text{ Lie} \cap B \text{ Truth})$$

Tool Name	The Rule	Example
The Magic Bridge	Moving a term across standard = flips its sign.	$X = Y - Z \implies X + Z = Y$
Keep-Change-Flip	To divide fractions, multiply by the reciprocal.	$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
Stack Eraser	If stacked fractions share a denominator, erase them.	$\frac{4/52}{12/52} \implies \frac{4}{12} = \frac{1}{3}$

5 Algebra & Arithmetic Essentials

6 Standard Deck Quick Reference

Total Cards = 52

Colors = 26 Red, 26 Black

Suits = 13 Spades (♠), 13 Clubs (♣), 13 Hearts (♥), 13 Diamonds (♦)

Face Cards = 12 total (4 Jacks, 4 Queens, 4 Kings)

Aces = 4 total